IEOR E4101 Probability, Statistics and Simulation — Practice Midterm (Alt B)

Instructor: (same as original)

Instructions: Show your work. Justify any formulas you use.

Problem 1 (Discrete probability)

You throw a fair 10-sided die (faces 1–10) ten times.

(a) What is the probability that all ten face values appear?

(b) What is the probability that one or more 10's appear?

Problem 2 (Bayes' rule)

A 100%-reliable sonogram reveals a couple will have **twin boys**. In the population, the proportion of **identical** twins is **25%**. For identical twins, it is equally likely to be two boys or two girls. For fraternal twins, the sexes are independent and equally likely. What is $P(\text{identical} \mid \text{twin boys})$?

Problem 3 (Scenario returns, correlation, portfolio risk)

Three equally likely scenarios: *strong, normal, weak*. Returns:

Scenario	Stock A	Stock B
Strong	22%	−12%
Normal	5%	7%
Weak	−8%	0%

(a) Compute $E[A]$ and $E[B]$.

(b) Compute $\mathrm{Corr}(A, B)$.

(c) For a **50%–50%** portfolio, compute the **expected return** and **standard deviation**.

Problem 4 (Overbooking)

No-show rate is **10%**. The plane holds **60** passengers; the airline sells **63** tickets. Let X be the number who show. Compute $P(X \leq 60)$ if $X \sim \mathrm{Bin}(63, 0.9)$.

Problem 5 (Records)

Let $X_1, X_2, \ldots$ be i.i.d. continuous. Say a **record high** occurs at time n if $X_n > \max(X_1, \ldots, X_{n-1})$. Define Y_n as the number of record highs by time n (with a record at time 1).

(a) Find $E[Y_n]$ for general n.

(b) Find $\mathrm{Var}(Y_n)$ for $n = 2$ and $n = 3$.

Problem 6 (Expected profit with a triangular density)

Weight X (oz) has density

$$f(x) = \begin{cases} x - 9, & 9 \le x \le 10, \\ 11 - x, & 10 \le x \le 11, \\ 0, & \text{otherwise.} \end{cases}$$

Selling price \$2.20; refund if $X < 9.40$ oz; cost $c(x) = x/18 + 0.40$. Compute the **expected profit**.